Assignment 13

Chapter 11

2. Let X and Y be two jointly distributed random variables with Y being continuous. The random variable Y is related to the random variable X through a conditional pdf $f(y|x)$ defined for all x (cf. Definition 10.22). Prove that $I(X;Y)$ is concave in $F(x)$.

3. Refer to Lemma 11.18 and prove that

$$\Pr\{(\mathbf{X}', \mathbf{Y}') \in \Psi^n_{[XY]\delta}\} \geq (1-\delta)2^{-n(I(X;Y)-\delta)}$$

   for n sufficiently large.

4. Show that the capacity of a continuous memoryless channel is not changed if (11.23) is replaced by

$$E\left[\frac{1}{n}\sum_{i=1}^{n}\kappa(x_i(W))\right] \leq P,$$

   i.e., the average input constraint is satisfied on the average by a randomly selected codeword instead of by every codeword in the codebook.